

1. Verification & Evidence Integrity

- **Calculate Hash Values** (MD5, SHA-1, SHA-256)
- Ensure hash of **original disk** = hash of **forensic image**
- Document chain of custody

2. Disk Information

- Drive model, serial number, size
- Partition details
- File system (NTFS, FAT32, EXT4)

3. File System Forensics

Recover deleted files

- Deleted documents, images, PDFs
- Deleted browsing history
- Deleted logs

File carving

- Recover files not in directory structure (.jpg, .png, .zip, .mp4, etc.)

Metadata analysis

- File creation/modification times
- Owner information
- Windows timestamp anomalies (Time Stamp Manipulation)

4. Operating System Artifacts

Windows Artifacts:

- **Registry Hives**
 - SAM (User accounts)

- SYSTEM (Mounted devices)
- SOFTWARE (Programs installed)
- NTUSER.dat (User activity)
- **Event logs**
 - Security logs (login attempts)
 - System logs (crashes, USB insert)
 - Application logs

OS System Files:

- Prefetch files → check program execution
- Recent files list (.lnk files)
- Thumbnail files (thumbcache)
- Recycle bin

5. Browser & Internet Activity

Check for each browser: Chrome, Edge, Firefox, Brave.

- Browsing history
- Download history
- Saved passwords
- Cookies/sessions
- Cache data
- Autofill data
- Extensions (malware or proxy)

6. USB & External Device Activity

- USB insertion/removal timestamps
- Device ID and manufacturer
- Files copied (if detectable)
- Suspicious data exfiltration

7. Installed Software & Logs

- Suspicious programs
- Remote access tools (AnyDesk, TeamViewer)
- Keyloggers / RATs
- Cracked software
- Batch scripts or .exe files created manually

8. Malware & Suspicious Activity

- Malware samples
- Suspicious DLLs
- Persistence mechanisms
 - Run keys
 - Scheduled tasks
 - Startup applications
- Hidden partitions
- Rootkits

9. Communication Evidence

- Emails (Outlook PST files)
- Chat files (WhatsApp Desktop, Telegram Desktop, Signal)
- Cloud sync logs (Google Drive, OneDrive)

10. Timeline Analysis

- Complete timeline of:
 - file changes
 - browser history
 - login sessions
 - USB activity

Magnet AXIOM / Autopsy / EnCase can auto-build timelines.

11. Images, Videos & Document Forensics

- Search for hidden or renamed images
- EXIF metadata
- Hash matching (CSAM detection / known malicious files)
- Steganography check

12. Final Validation

- Recalculate hashes
- Ensure evidence integrity
- Compare with initial hash